

INMAS: Internship Network in the Mathematical Sciences

Year 3 Evaluation Report

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Introduction

The Internship Network in the Mathematical Sciences (INMAS), a five-year project funded by the National Science Foundation (NSF), is co-directed by Fadil Santosa (Johns Hopkins University) and Richard Laugesen (University of Illinois, Urbana-Champaign). The goals of this project are to:

- Provide high quality training and internship experiences that broaden career opportunities for students in the Mathematical Sciences.
- Bring powerful computational and modeling tools to bear on industry's most pressing problems.
- Build mutually beneficial connections between industries and mathematics departments.
- Support systems change within mathematics departments in support of expanded opportunities for students and industry partners.

The major activities planned to accomplish these goals are the following.

- Launch regional networks, with hubs at Johns Hopkins University and the University of Illinois and nodes at graduate level institutions in neighboring states.
- Prepare students for the internships through technical and professional skills training.
- Seek out challenges in industry and government organizations that can be tackled by Mathematical Sciences techniques.
- Create internship projects at those host organizations, and match with students in the network.

In the third year of this program, students who were selected to participate in INMAS attended four technical skills weekend workshops covering Python programming, statistical methods using Python, machine learning, and a final 'capstone' team project. Additionally, students attended online sessions to develop professional skills (resume development, elevator pitches, and behavioral interviews). Two of the weekend workshops were at the hub locations (Johns Hopkins University and University of Illinois Urbana-Champaign); students gathered at their local node universities for the remaining workshops.

In the third year of the program, 91 of the 141 students who applied to INMAS were accepted. Fifty-one students affiliated with the Midwest hub and 34 students affiliated with the Mid-Atlantic hub completed the technical and professional development training workshops. Forty-one students (32 Midwest hub and 9 Mid-Atlantic hub students) were placed in INMAS internships. The remaining students applied for and secured other internships on their own (using skills developed during INMAS), selected to focus on their research projects over the summer, or deferred the internship by one year due to advisor or personal reasons.

In selecting companies to participate and host a graduate mathematics intern, INMAS sought companies that had projects involving a high level of intellectual sophistication including some amount of mathematical or statistical reasoning. In particular, INMAS looked for places where mathematically-trained talent could make a difference and where the company did not have a history of engaging mathematical scientists. In the Mid-Atlantic, small-to-medium sized companies were targeted because we wanted to maximize potential impact to the company. The methods by which candidate host companies were identified included:

- recommendations from node university personnel in academic units and corporate relations offices,
- companies with whom the PIs have had contact in previous years,
- companies identified as holding federal Small Business Innovation Research (SBIR) grants, and
- companies located in university-affiliated research parks/business incubators.

Companies that seemed promising were then contacted by email. If they responded positively, a Zoom call was arranged with their staff to understand better their business challenges, and to learn more about potential projects for interns.

Evaluation Design

The goal of the evaluation was to track successes and challenges of the project, paying particular attention to any adaptations made based on the previous year, and to provide program staff with data, observations, and recommendations for continued improvements. The following four overarching evaluation questions continued to guide the inquiry.

1. How did students experience the INMAS program activities?
2. To what extent did the technical and professional development trainings affect students' awareness, skills, and confidence?
3. How did students and industry partners experience the internship matching process including preparation, opportunities, and selection?
4. What was the overall experience of the internship including fit of student skill/interest/knowledge with industry and benefits to students and industry partners?

Evaluation Activities

The evaluation activities for the third year included the following:

- At the beginning of the program, an online pre-training survey was emailed to 91 students in October 2022 to learn about their technical and professional skill levels prior to any INMAS workshops as well as perceptions about industry opportunities. Eighty-eight students completed the pre-training survey.
- Following each weekend workshop, a survey link was emailed to students as follows:
 - October 2022: Python Programming (75 respondents)
 - December 2022: Statistical Methods using Python (64 respondents)
 - January 2023: Machine Learning (48 respondents)
 - February 2023: Group Project (66 respondents)
- The February 2023 survey also included a post-training section of questions (to match the pre-training survey taken in October 2022); 66 students completed that survey.
- A brief follow-up survey was sent to students who had participated in the first two years of the program (44 respondents).
- Following the internship experiences, interviews were conducted in September and October 2023 with a select number of industry staff and students. Thirty people participated in the interviews, split evenly between students and company staff.

Survey data included descriptive statistics as well as qualitative responses to the open-ended questions. All interviews were conducted via Zoom and were recorded with participants' permission. Interview recordings were transcribed and uploaded into NVivo, a program that supports qualitative analysis. Direct quotations from the interviews are used to illustrate thematic points throughout this report. At each round of data collection, findings were shared with INMAS staff to support ongoing program improvements.

Evaluation Findings

The key evaluation findings followed by supporting evidence from the data are presented in four areas: technical and professional development workshops, student perceptions of industry careers, internship experiences, and recommendations from industry staff and students.

Technical and Professional Development Workshops

As noted previously, four technical workshops and three professional development sessions (resumes, elevator pitches/LinkedIn, and behavioral interviews) were conducted during the fall and winter, 2022-23. The technical topics included Python Programming, Statistical Methods, and Machine Learning. Workshops were developed by INMAS staff and TAs, and by externally contracted instructors, building on the experiences of the first two years of the program.

Key Findings: Workshops

Student Gains in Technical Skills and Confidence

- Students had varied levels of experience prior to the workshops ranging from little experience to proficiency. Post-training survey responses show a growth in skills in all three areas (Python, statistical methods, and machine learning).

Student Gains in Professional Development

- Students reported an increase in confidence in all four areas of professional development (resume preparation, networking, interviewing, and teamwork).
- Students noted the importance of teamwork.

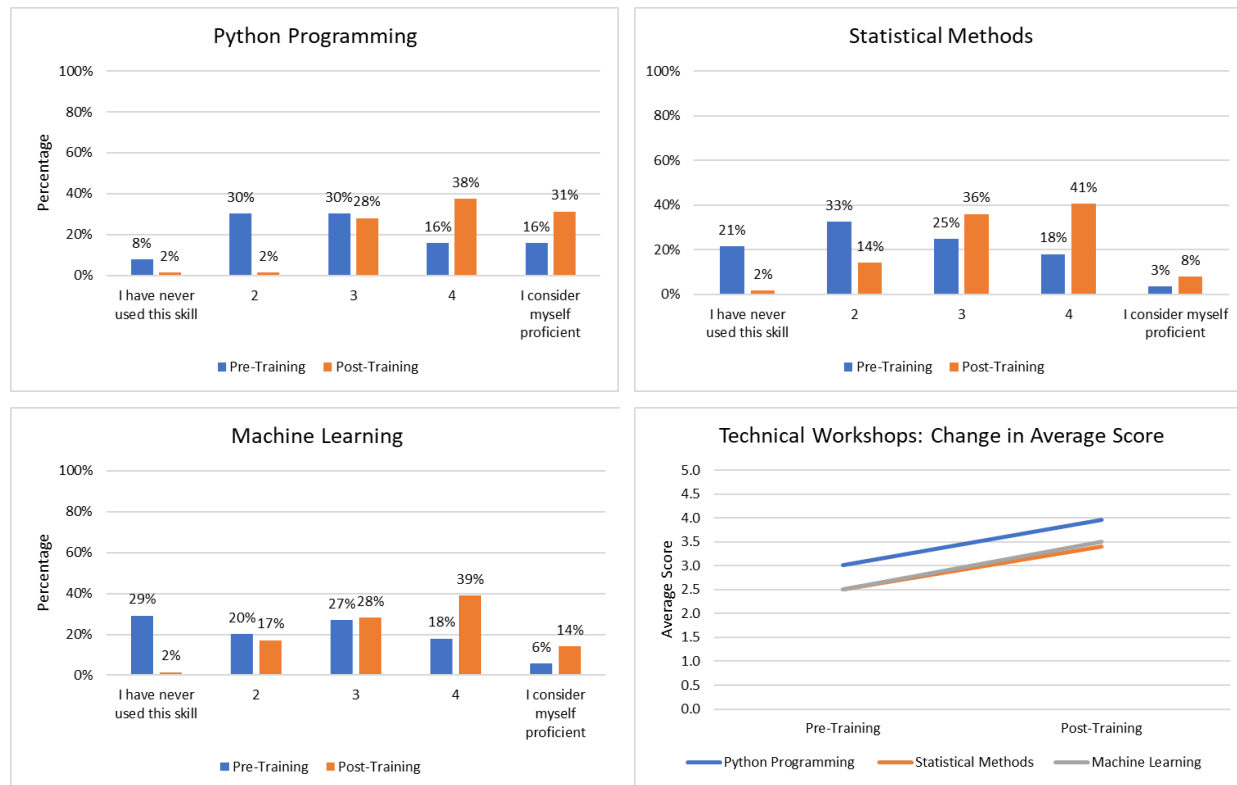
Student Recommendations for Workshops

- Students recommended providing more guidance in teamwork, allowing choices in attending workshops to fit training and interest, and focusing more on the final project (as it is often one of the few examples of work a student has to share with a potential employer).

Supporting Evidence from Pre- and Post-Workshop Surveys and Interviews

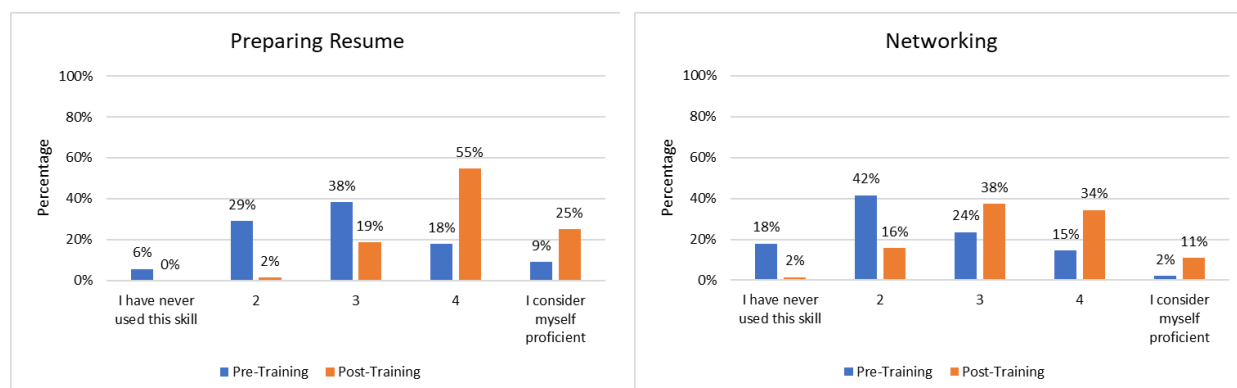
The following are data from the pre- and post-workshop surveys. Invitations were emailed to 91 students; 88 students completed the pre-survey and 66 students completed the post-survey. Students were asked to rate their level of knowledge on a 1-5 scale: I have never used this skill (1) to I am proficient (5).

Pre- and Post-Training Survey Results: How would you rate your proficiency in the following technical skills?



Professional development activities focused on resume and interviewing preparation along with teamwork (via group projects). The following are the survey results for these areas.

Pre- and Post-Training Survey Results: How would you rate your proficiency in the following professional skills?





Student Reflections on Workshop Experiences

Students were asked in the post-workshop survey what they found most valuable about the workshop experience. The following key examples of their open-ended responses from the surveys.

- The importance of teamwork was noted by a number of students. For example, one student noted, “Teamwork is the key to success. I really enjoyed the teamwork on the last workshop. It led to various ‘out-of-the-box’ ideas to approach problems and solve them.” Another student wrote, “Teamwork can sometimes be rocky, but effective communication is the key to making progress.”
- Students noted particular technical skills that they learned:
 - Importance of python proficiency
 - “INMAS sessions allowed me to get started with Python, and learn how to implement in Python nontrivial tasks for which I would have used MATLAB in the past. I now have good resources to deepen my knowledge of it by myself.”
 - “The most important takeaway was how there was such depth to the material and that I felt like I need lots of practice to further develop my programming, statistical analysis and machine learning skills.”
 - “Aside from learning new programming skills, the workshops were helpful in teaching us a variety of different tools in ML and Stats that we can use on our own. I was also able to practice how to debug which I think is essential. I especially liked that during the last workshop we were able to download data and really figure out how to clean the data and make it usable in order to analyze it. Usually when I work with data, professors organize the data so that it's ready to use.”

- “I learned a lot about programming and statistics. The most important skill I would say is the ability and mindset of problem solving of industry-related problems.”
- “I can do much more than I thought, by breaking these problems into smaller problems.”
- Students valued the professional development activities:
 - “Resume preparation and providing us internship proposals from companies.”
 - “For me, the most valuable sessions were those related to resume prep, interviewing, and networking.”
 - “I found that it's useful to follow through on ideas in a team even if I don't completely agree with them or feel they will lead to fruitful results. It's also sometimes useful to talk with mentors with more experience outside the group for advice.”
- Additional comments from students:
 - “An industry career might actually be intellectually rewarding!”
 - “If it wasn't for INMAS, I would not have applied for internships simply because I did not know what to expect outside of academia.”

Student Recommendations for Workshops

Students shared recommendations for future iterations of the program. These suggestions were given in both the post-training survey and the interviews with selected students following the internship. In the open-ended responses from the survey, students noted their preference for in-person workshops while also understanding the logistical constraints, the need for additional guidance on working in a team, and an interest in having more optional datasets and projects throughout the program. During the interviews, students underscored the learning curve for those without prior programming experience and reiterated some of suggestion given in the workshop surveys. The following are quotations from the interviews.

“I found the learning curve for the programming workshops to be pretty steep. They gave us a lot of materials for the programming workshops to the point where I didn't finish all of it just because there's not enough time to finish all of it inside the workshop. I know for some of my friends [who didn't have programming experience] it was tough.”

“I think, from my perspective, not all of the workshops were explicitly applicable. I think they made me a better mathematician overall, so I think that they were helpful but some of them, for the amount of time that we were devoting towards it, weren't as beneficial. I don't know if there's a way that they could say, ‘You need to attend three workshops and choose the three that fit your growth or fit your particular field and use those as beneficial,’ because I think there were two of the four that weren't super applicable to me.”

“I think the biggest recommendation I have is for them to be more intentional about how they want and expect us to work in groups in those workshops. I think that was one of the reasons I felt unprepared is because we are all graduate students, and I feel like at this point, most of us know how to effectively work in groups but we were not effectively working in groups. And so maybe some more guidance, at least in the beginning to get people comfortable with doing that.”

“There was a final project in the workshops. I think maybe they can make it into a bigger, longer project, because I think that's what companies make their decisions based on, especially since we usually don't have work experience.”

Student Perceptions of Industry Careers

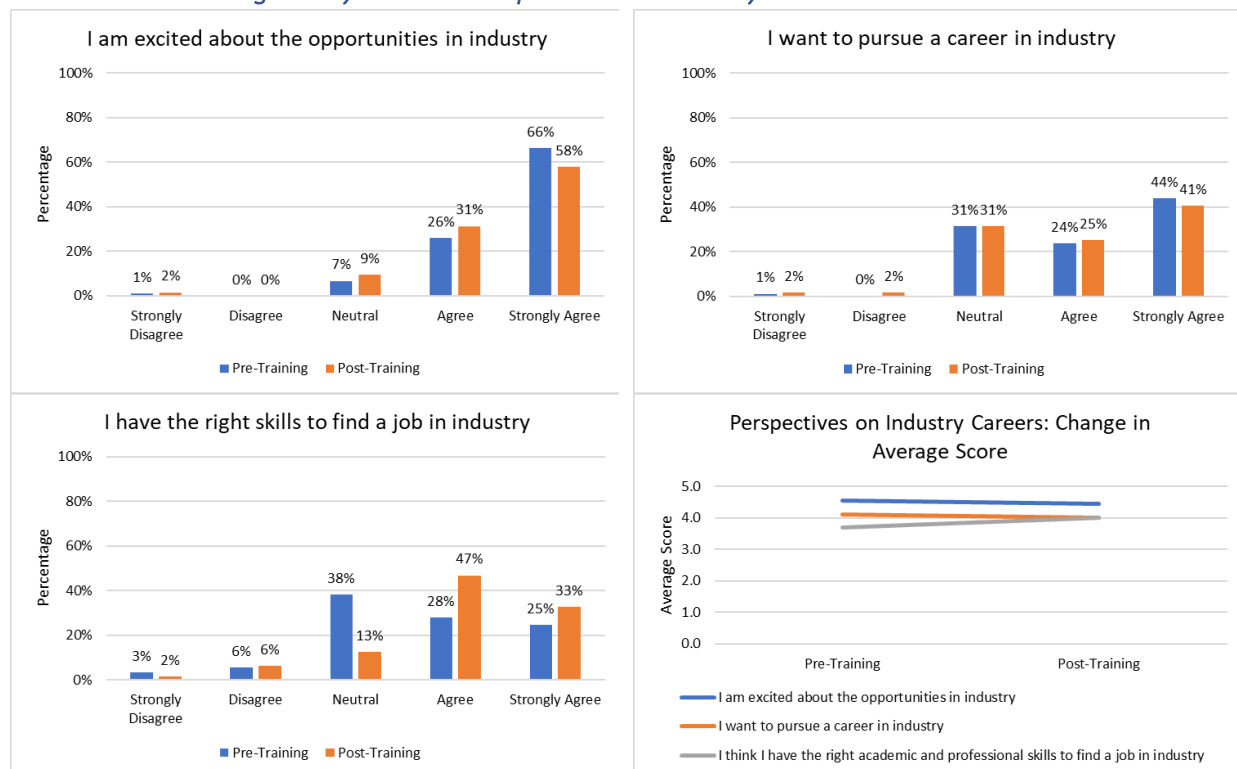
Students' perceptions of industry careers were assessed at the beginning of the program and at the end of the technical and professional development workshops via the pre- and post-workshop surveys. Selected students were also interviewed at the completion of their internship experience.

Key Findings: Perceptions of Industry Careers

- Overall perceptions about industry careers did not change as much by the end of the workshops; many questions about working in industry still remained.
- All of the students interviewed after the internship experience shared that their understanding of the opportunities in industry careers had grown or had been clarified. Many indicated that the experience confirmed their choice in career direction. Eight of the 15 students who were interviewed said they already were leaning towards an industry career and the internship confirmed that decision. Two students planned on continuing an academic career path and four students were still undecided. One student felt that he was able to do both by accepting a teaching position at a college that would allow him to continue research projects in the summer.

Supporting Evidence from Pre- and Post-Workshop Surveys

Pre- and Post-Training Survey Results: Perspectives on Industry Careers



Supporting Evidence from Interviews with Students

While the internship experience expanded their awareness of the possibilities in industry, most students interviewed this year noted that the internship confirmed the direction they thought they would pursue for their career path. For example, 8 of the 15 students said they already were leaning towards an industry career and the internship confirmed that decision. Two students planned on continuing an academic career path and four students were still on the fence. One student felt that he was able to do both by accepting a teaching position at a college that would allow him to continue research projects in the summer. The following quotations illustrate these themes.

“This internship helped me as well because it helped me to see more opportunities. I didn't know a lot of opportunities in that field existed before, but after working with them I feel like there are a lot of work that could be done, and there are open positions as well.”

“I'd already been thinking that the geographic challenges of jumping postdoc to postdoc would not make sense for me and my partner. I was already thinking about other things but I think I was very much not knowing what that would look like. I think the internship made me feel like I'm less nervous about that now because I was able to jump into this different company, learn these different things, mostly adjust to this different way of interacting with people, and then, able to actually create something that I'm now still finishing up.”

“Sure, academia and industry go hand in hand but the kind of problem I get to work on in my academic setting, these are not bounded by what a client needs or not what the current technology can do and cannot do. So it's kind of like an open playground, I would say. While I'm implementing something for my company, it's completely different. I have to think about what the company needs. I have to worry about, ‘Well, we don't have the hardware to implement that because it's going to take so much computing power. We just don't have it.’ And all these other things that I just feel that in an academic setting I get to work on a lot of harder problems with a lot more freedom that I don't think industry can match.”

“I think the internship opened my eyes, the possibility of non-trivial work outside academia. So that's one. And then moving forward, I think I told myself that, ‘Actually, this is not that bad. This is still interesting work.’ ... People told me, ‘Just apply to both make decisions later based on what you got.’ But I would say that if I'm doing postdoc, I would like to pick my problems that probably slightly closer to applications. That's probably the concrete answer that I can say.”

Internship Experience

As noted earlier, INMAS staff used criteria to select companies with the greatest opportunities to utilize and benefit from interns with high-level mathematical training. INMAS staff created online internship listings for the identified opportunities. INMAS students were able to see all positions from both the Midwest and the Mid-Atlantic hubs, and select internships that fit their location, research interests and skills, and their preferred area of industry. The Project Developers at the hubs, who communicated directly with industry partners about participating, also met with each student to learn more about their interest and skills and sought to beneficially match students with each internship opportunity. In addition, INMAS staff hosted an online networking event for both students and industry partners.

The finding and the evidence presented in the following sections is based on 28 interviews conducted with participating industry partners (15) and student interns (15). Note: two of the interviews with industry partners were conducted with two people from the company.

Key Findings: Internship Experience

INMAS Process

- All industry partners and students reported that the initial phases of the process that led to each internship offer (including identifying potential companies and projects, assessing student interest and preparation, facilitating students' application via resumes and interviews) was conducted well. Many noted that the process was surprisingly straightforward and easy.
- Timing was one of the few issues that was mentioned. Some industry partners noted that they were "late" in learning about the INMAS partnership so the pool of available students was smaller. Students mentioned that receiving offers in late May was challenging, especially for international students.
- Three-quarters of the partners who participated in the interviews said that the \$5000 stipend from INMAS was important or very important in their considerations of going forward with an intern. All industry partners were interested in participating again.

Company Motivation for Participation

- Industry partners identified three main motivations for participating: start-ups or small companies often have a need for more staff in the immediate term, many want to give back to the field or support young researchers, and internships provide a useful way to vet and hire new staff.

Academia vs. Industry Culture

- This year's interviews revealed a more nuanced consideration of the differences between academia and industry. In particular, a number of students observed that their first experience in industry only offered a singular window to what is obviously a diverse field of experiences in industry.
- Both students and industry partners discussed how the characteristics of initiative and self-direction along with the challenges of real-life data might be viewed or encouraged differently within these contexts.

Overall Internship Outcomes

- Most students reported that they were satisfied with the final product of their internship.
- Most industry partners reported that they were satisfied or extremely satisfied with the outcome of the internship and final project.

Recommendations

- Industry partners offered recommendations on preparing students for the internships, clarifying some of the INMAS processes, and building more collaborations with industry. Industry staff also had some recommendations for how companies can improve their own preparation for hosting an intern.
- Students offered recommendations on increasing opportunities for students to interact with and learn from industry partners and clarifying some of the INMAS processes.

Supporting Evidence from Industry Partner and Student Interviews

Student Feedback on INMAS Process

Overall, students who were interviewed reported a high level of satisfaction with the process. They noted the ease of using the online job postings, and the support and guidance from INMAS staff. A few students also mentioned the online networking event hosted by INMAS.

“It was a really seamless process, I thought. It was very nice to have the board of internships that INMAS put together [online], so we could just, in one place, look at all the internships. You could see what the requirements were, whether you needed to be a citizen, what the skills were. There was a PDF that I think the company sent to INMAS just to say, this is the project we're interested in working on. So it was really nice to have all that information collected into one place as compared to if you were to do something on your own, you'd have to do a lot more research and get it from 50 different company websites instead of it all being collected in one place. So that was very helpful.”

“My experience was actually fantastic. After we had the initial meeting where we met a lot of these representatives at the online meeting, they were pretty receptive to what I was able to offer. So when I applied to these places, they quickly brought me in for an interview, and I think I had about five interviews total.”

“I think it was pretty smooth, very straightforward. All the information was there, and picking which ones I wanted to apply to was easy. The application process, I guess, was pretty simple and didn't feel overwhelming, that was one of the things I was worried about. I was only interviewed by one company, so the choice of internship was pretty easy as well. ... The company that was my first choice was the one I was interviewed by.”

Students also mentioned the importance of direct support and guidance from INMAS staff.

“I remember in December of last year, I was at the grocery store and I got an email from [INMAS Project Developer] saying, ‘Hey, there's this internship [at a particular company] that looks like it would really fit your interest.’ He had interviewed me before and talked to me about what I was interested in. ... And so that night I sent out my application through the monday.com board, and I think within a week or two, they responded to me and said, ‘Hey, we really like your resume and we want to interview you.’ ... I'm incredibly grateful for everything that INMAS did, especially to [the INMAS Project Developer] with the fact that it was very helpful that he had met with everybody at the beginning just to see what our interests were and the fact that he remembered what my interest was. So when he saw the internship pop up for this company, he specifically contacted me and said, ‘This is the one. Go check it out.’ That email was life-changing, quite literally.”

“He [INMAS Project Developer] encouraged me to try and network and find additional companies that are open to the idea of joining the INMAS group, the companies that are connected and affiliated with INMAS. And so, I did. ... It worked out really great. I got to build some great ties, and I got to work in areas that were my field of expertise, and in some that

weren't. So it was kind of a great collaboration. INMAS provided just the structure and framework to be able to set this thing going. For me, it was an amazing opportunity of networking and building relations.”

Industry Partner Feedback on INMAS Process

All of the industry partners who participated in interviews said the process of communicating with INMAS and selecting an intern for the summer was smooth and straightforward. A few Industry partners noted that they had previously participated in INMAS, and continued to find it beneficial. One returning company shared that they felt the quality of the pool of candidates was higher than in previous years.

“The resumes were well aligned with the objective of the summer research internship. There was a good variety of candidates – some very high-quality candidates, some in the middle of the pack, some not so high quality. But overall, I think I had enough quality candidates to choose from.”

“Everybody INMAS sent us was good. I mean the only filtering criteria we had for the type of contracts we have is US citizenship. So that’s kind of number one filter from the list that got applied. I mean, everybody else is a good hire. That’s how we looked at the resumes.”

“By the time I found out about it [INMAS], I think already selections were in progress. But I reached out to [the Program Developer] and then he was able to find a couple of resumes. I reviewed them and they looked good. The thing is they looked a little bit more theoretical math than the applied mathematics that I need. At the time, I was a little concerned since I was looking for more engineering oriented than just pure math. Our work is very software-oriented. So having to write code, Python, use AI/ML tools. ... But I reached out to both candidates that I had resumes for. I did get contacted by the gentleman who ended up working with us, and he was amazing. He was so great to work with. Just an excellent attitude. ...and he knew a lot more about software than showed up on his resume.”

Within the range of positive experiences, there were a few points of consideration that industry partners also shared. For some, starting the applicant reviewing process late resulted with fewer choices that did not turn out as well as the situation described in the previous quotation. Others noted that certain important skills were not as easily captured in a resume.

“I think we were at the tail end of the process. I think we only got the resume of two people. ... So, I don’t think that’s quite the range we would have like to have seen. We were trying to follow our standard process for hiring but we knew this was a subsidized position. So we knew that if they were on the edge, I think we’d still be willing to take them. I think that was the case here, that [the student] definitely wasn’t passing our interview with flying colors, but we know this is a training program, this is an internship... Also, something we didn’t do, which we probably should have done, again, because we thought it was an internship, we never contacted the student’s advisor. We would normally check references and follow up with their advisor or whoever, especially for a PhD candidate, we didn’t do that in this case and we definitely should have.”

“I think there was definitely really smart people in the pool that I was given, but I was really looking for that person that had something on the resume that was going to tell me, ‘I can

collaborate. I can work with other people.’ I didn’t see that in several of the submissions, and that’s hard to articulate on the resume too sometimes. They could be very good people, people, but we just can’t see that on the resume.”

Importance of Stipend

Three-quarters of the partners who participated in the interviews said that the \$5000 stipend from INMAS was important or very important in their considerations of going forward with an intern. For some companies, especially the smaller start-ups, the additional funding allowed them to offer an intern a competitive salary. For those who said the money itself was not that important (or was even challenging to deal with via the company’s internal financial policies), they often noted other benefits of partnering with INMAS. For example, one staff member said, “But having that platform actually, the ability to select the students from getting the student talent pool, that is the main key. The stipend part is not that important.”

Company Interest in Future Participation

As in previous years, all of the industry partners who participated in the interviews indicated that they were interested in participating in the INMAS program again. The only caveat was that it would depend on what was happening in the company, in terms of available resources and time as well as wanting to have a great project for an intern to work on.

Company Motivation for Internships

Motivation for hosting internships fell into three main themes:

- ***Staff of small start-up companies reported a need for extra staff that they could tap into quickly.***

“That’s why I loved your program and I’m looking into getting more people from you this year if you have anybody because we just finalized the job description right now in computational mathematician, so we’re looking for hiring more. If you guys have more people, we’ll interview them.”

- ***Industry partners were champions of supporting the next generation of researchers and data scientists, even sometimes pushing the agenda within their own companies.***

“I always go back to my own PhD days. I wanted someone who could have shown me how industry works. So I want actually just to give back, as much time as I can. Our company owns this company that specialize in statistical math. We have all the resource internally and I actually got challenged quite a bit about why we are participating in this [the INMAS program], when we already have resources in house. It’s always just giving back and actually helping students.”

“We want to see people be successful, we want to see them grow, and we want to get value while they’re with us. And so we put a lot of intentionality into our interns’ overall intern strategy to make sure that they’re successful.”

- ***Industry partners reported that internships were part of their hiring pipeline.***

“It [hosting an INMAS intern] works well because we looking for fresh ideas, vibrant, energetic coworkers. We’re looking to develop talents, tech talent development. We’re looking to fill a

pipeline. So, I think some of these interns that we hired, will probably work for us full-time down the line. So for us is just part of the holistic recruiting and personnel development efforts that we had.”

Differences between Academic and Industry Culture

This year’s interviews revealed a more nuanced consideration of the differences between academia and industry. In particular, a number of students observed that their first experience in industry only offered a singular window to what is obviously a diverse field of experiences in industry. However, both students and industry partners discussed how the characteristics of initiative and self-direction along with the challenges of real-life data might be viewed or encouraged differently within these contexts.

Student Intern Perception of Industry Culture

When students were asked about how they experienced the shift from academic to industry culture, their responses fell into the following themes:

- ***The project and the company resembled academic research more than students expected.***

“That [adjusting to industry culture] particularly was not as much of a problem simply because the nature of this internship. My project was developing an internal tool for the company. And it was based on a lot of research. Every day I would read three research papers, put down what's important to keep in mind while implementing a certain thing and what doesn't work, what other people have found to work. So, it was basically like doing research in, let's say, as you do in a computer science department.”
- ***The industry culture students experienced seemed less hierarchical and more supportive of a work-life balance.***

“It was a good experience. I really enjoyed my coworkers. I worked with a lot of the tech team, so it was a lot of just working with guys but I'm pretty used to that in math, so that's okay. But it was fun. It was a good environment. Obviously, it felt different. I would say it felt less toxic at times. It was not as competitive [as academia].”

“It was very collaborative, so I liked it a lot.”
- ***Some of what students had heard about industry did not occur.***

“One thing that had actually worked out in my favor is that I heard that a lot of companies did daily standups where you would check in with your supervisor every single day. My company actually did something that I prefer, which is a weekly standup. ... I came in with the expectation that, ‘Okay, I'm going to have meetings because every company has a whole bunch of meetings and that's just how it is.’ It actually turned out that it wasn't like that. It's one meeting a week or maybe two meetings a week if you're a part of a couple of teams simultaneously, just to get a broad check-in, no more than 30 minutes, and that was that.”
- ***A range of decision-making processes were experienced: some students experienced a quicker pace with decisions and others experienced more circuitous decision-making shaped by internal dynamics.***

“When I got into my internship, I recognized that right away everything was faster. You would solve a problem and people would be less interested in the very minute technical details, and they would be more interested in the big picture, ‘Does this work? Does it capture what we're trying to do? Does it help the customer?’ These sorts of things. That sort of thinking is what I'm really interested in because I want to get to the technical heart of whatever the problem is, but I also want to just get things to work and then move on. ... It was really nice in the transition to industry to be able to spend more time on the big picture, thinking of the ultimate goal of what we're trying to do, rather than being stuck in the weeds trying to figure out this little detail because somebody's going to ask me a question and if I answer wrong, I don't get a degree.”

“I spent a whole week doing part of the code that was another feature and it got rejected during the design meetings. Most of the challenge was having everyone in the same mindset.”

Students noted that they realized that their internship experience – for many, their first – was only a singular experience of that one company rather than ‘industry’ overall. Some students noted that they were interested in applying for more internships in the future to broaden their experiences.

Initiative, Self-Direction and Ownership [Reflections from both Industry Partners and Students]

All of the industry partners who were interviewed talked about the importance of initiative and self-direction for the interns and staff who work in their companies. As one staff person observed,

“That's pretty critical for startups in innovation because if we hire people who need to be super managed, we couldn't function. We're just too small right now. When we're super big and we've got 500 people and we're just growing, crushing it, we can invest in our own training and have a serious recruiting program where we take early career hires. But for us, this [INMAS] is really helpful, because these students come in with that kind of creative background and knowledge of data science that makes them relatively productive early term, and then, valuable long term.”

In a number of the internships, industry partners were pleased with the self-direction that they saw the interns taking.

“I was kind of wondering, because I didn't really hear too much, if we were leaving him [the INMAS intern] out to dry or something. Then, he would just keep producing. He showed a real interest, especially in the second half. We were working on something – actually, it is kind of money-ballish, kind of cool. It's at the intersection of finance and data science. He was really instrumental in helping us do some correlations, showing that the input data that we have was highly correlated to certain price spikes. ... Then the intern took that further and showed some real interest in it and was, by the end, teaching us the script, teaching us maybe where we should go next.”

“He really worked well with the team, collaborated, and he wanted to learn how can I use all these complex skills I'm using in the real world? He wanted to see how can we fit that in. Not all PhDs are like that, but he was very much like that. It worked out really well.”

However, in four of the industry partner interviews, staff reported struggling with the intern's lack of self-direction or initiative. In one case, the industry partner reported that the project was handled well, however, he still felt compelled to give the intern some personal feedback.

"Because in my setting, he was a bit timid while he shouldn't have been so because in my opinion, he is trained enough. ... It just takes time and he just needs to be more sure in himself in finding future projects. He's skilled enough and he's trained enough. Just a question of time to get more knowledge in data science." [industry staff]

In one situation, the industry partner had two INMAS interns. He reported that he was pleased with how the projects turned out but noted the difference between the two interns.

"Some people have this capacity to listen and get the picture, and be more flexible, and try to understand what is going on, and try to come out with ideas. One of them [the interns] was more like this. The other, it was more, I will say more spoon-feeding. It was harder for that intern to be in an unpredictable situation. Startups are totally unpredictable. The intern had a harder time to adjust to these realities, of changing. ... However, I think that they were able to do some good progress. ... they end up both kind of working with the other, with the one more support to the other." [industry staff]

In two of the situations, the industry partners expressed disappointment or frustration with the intern's lack of initiative.

"I had tried almost for first eight weeks or something [reaching out to the intern for updates] and there was no feedback or output coming. It was only one-direction flow. We were meeting once or twice a week in the beginning. Then actually, I felt like that there is no engagement so I actually just stepped away. [This was markedly different than the employer's experience in the previous two years of participating in INMAS, which led to his next reflection.] But I think, based on my own experience, I would give an opportunity to students who are in their final years rather than first or second year. I think they are really ready to stop, think or ask themselves the questions. 'How they can connect their knowledge that they have gathered from academics to the real-world applications, whether it may be in academy or also in the industry.'" [industry staff]

The next few paragraphs are quotations from the other industry partner who found the lack of initiative challenging.

"I guess two things, the student's specific mathematical background, we didn't really understand too well. It's differential geometries but in spaces that don't exist. I think when the student thinks about these deep learning models and imaging, that's the closest connection they can make. But it's not in any way a direct one. ... we did try to engage the student on several conversations about specifically what has they done and how that might apply. And nothing, as in I think the student had made the decision that it's not applicable.

But then it's the other set of skills, the set of skills that you need to have for any sort of knowledge worker, any sort of proactive person in any sort of professional context. We both come from a consulting background where those skills are emphasized a lot. We were hoping that they'd still have all those skills because they got a PhD. And it's a math PhD, too. I think

that's where we were the most surprised that the student didn't display in these contexts, basic problem-solving skills or demonstrate taking ownership of problems. I think a lot falls under this broad level of problem solving, and taking personal responsibility, taking ownership, being proactive, all of these things kind of wrapped up together.

What are some soft skills to have? ... how should you do just a quick presentation? What is that like? What is a work environment like? What is expected of you? What are the ranges of things that can be expected of you? ... We would expect people coming out of PhD program never, ever to show an unlabeled graph, for example. And same with you're going in for a meeting with your professor, it's your job to set the agenda for that. You are going into that meeting trying to get something out.

And so again, disappointed a little bit, and these are standard professional skills. When you're coming into a meeting, you're not just there to talk. If you're going to be presenting slides, you can't just jump into the detail. It always needs to have some sort of structure to it. ...This student didn't have these, and these are easy things to do, but super frustrating on our end when they don't have them.” [industry staff]

Real-life Data and Decision-making

Adjacent to taking initiative and being self-directed, industry partners and students talked about the difference in working with real-life data and decision-making amidst imperfect situations.

“We're not as concerned with people being 100% right all the time. Where in PhD-land, you want to be right because it's how you get your reputation. We just want people to get to the right answer quick. That's the magic. ...we're in the game, where we're trying to move fast, we're not as concerned with being right all the time. We'll be wrong, get some feedback, and the next time, we'll be right.” [industry staff]

“I would say the big difference is the problem statement in academia is well-defined. In industry, the problem statement is oftentimes aspirational in a desired outcome. So you really have to build the problem statement and that understanding. On the tactical side, data is given to data scientists in an academic setting that's perfectly built for data science. And the reality of industry is the only people with good data are academics. I've worked for Fortune 100 companies all the way down to tiny consultancies and they all have terrible data. So that is a real struggle for a lot of interns, is that realization that a lot of your work is not done doing data science, it's getting things to the point where you can do data science.” [industry staff]

“They all came in very confident in their ability to write SQL and work with databases, and none of them were up to the task. Because they were seemingly given a bit of a crash course in SQL, but again, real world data is much more intricate and much more involved. And it was something we worked through. Not a complaint, but I hadn't planned on spending as much time focusing on that area of growth for them as I had more on the specific data science side. But to be successful on the data science side, there was a lot more work that was done than I planned on the SQL side. ... To write the SQL in the academic setting, they're given a CSV file, all the data's there, it's all curated, it's all organized. Where our data is spread out over dozens of tables with varying levels of data quality because humans are terrible at data. And so being able

to dig in past that really simple, 'I want this data from this place,' it was something that I felt like was a struggle for all of them." [industry staff]

"I was given a very big data set and I need to fit all of them into some model. ... So I worked on that at the very first week. I thought, 'Okay, this should be fine,' but the result is actually not really good. I needed to first do some data cleaning and then fit into the model so I could get a better result. But I didn't do that. I just finished a part of the dataset. I asked my supervisor, 'Do you think this is good enough?' And he said, 'Ah, maybe you should...' I was super confused at that time because it doesn't fit well. It should. And the shape of the data, it looks like it should fit into the model, but I just keep working on myself and I didn't really ask for help. And I was kind of anxious because I think somewhere I did something wrong, but I just don't know it. So before that meeting, I was kind of super anxious and after we talked, my supervisor just said, 'Okay, maybe you should try to do the data cleaning at the beginning, then you will have a better result.' After that, I learned that if I have something that looks unreasonable, I should just email him and ask for some help. Yeah, don't wait and work on myself. Plus I need to clean the data." [student]

"I think this is a common theme that was there for many of us. Because many of us were coming from a theoretical background. Initially, I had a hard time kind of recognizing what the expectations are and how much time and effort I'm expected to spend if I'm stuck with something or where to cut corners and where not to cut corners. So that was, I think, the most challenging aspect. For example, I barely made any progress in my first five weeks because I was too bogged down with small details. I do mathematics, so any small incompleteness kind of bothers me so I was trying to fix everything. But by that end of my fifth week, I had a talk with my supervisor and I realized that, 'Okay, this is not going anywhere. I need to change gears.' So basically, the remaining six weeks was completely different and I was able to basically finish the project. I would say that was the most challenging thing – changing the mind state from an academic setting where you need to be perfect into a company setting where you need to deliver some stuff within a certain time." [student]

Teamwork and Supervision

Based on the students and staff who were interviewed this year, it seemed that many interns worked in small teams or primarily with their direct supervisor for most of the internship.

"I thought I would work with a group but actually I didn't. I just worked with my supervisor. But I would say he's super helpful. If I have any questions, I just emailed him because they are actually doing something related to chemical reaction and I don't have any chemical background. So sometimes I'm pretty sure I asked several stupid questions about the background because I kind of need to understand it to do the follow-ups. ... He was super, always encouraged me and super helpful, super patient." [student]

Additionally, most students noted that they met daily or weekly with their supervisor or the person working most closely on the project with them. However, a couple of students noted their confusion or lack of understanding as to how it was supposed to work.

"I was not sure that's how it worked because we did not have any days or date planned for the meetings. It was totally onto him when he wants to meet and what time he meets. He just shoots me an email and I have to be there." [student]

Industry partners also noted that at times they struggled with supervision or differing perspectives and expectations. One staff member noted that with one of the interns in particular, he wondered about productivity and commitment.

"How we should supervise timing and effort in the students? Because they are not a regular employees; it's just 8 weeks. We want to keep the productivity, and the engagement, and a positive environment. I always try to do this. But at the same time, when I appear nine o'clock and nobody in the office... nobody in the office until 11." [industry staff]

Overall Project Outcome

A majority of both industry staff and students said that they were satisfied or extremely satisfied with the overall project completed during the summer internship.

"Very positive, overall. They're all smarter than I'll ever be, which is awesome, and so I think eager to learn. Interns are weird in the data science, through INMAS or otherwise, a lot of people have ideas on what they want to do, but the nuts and bolt data science and industry is very different than academic data science, and it's a very different experience for a lot of folks coming in. I feel like they all did a great job this past summer of really stepping up and digging in and understanding more than just the data, but the overall business reason behind the work we're doing." [industry staff]

"The intern showed very, very good skills both in programming and he also performed independent studies by finding what others did previously in this area. And he identified weak points and also strong points of the models which they tried to apply. And he tried to modify their models and to apply to our own data. That's exactly what we expected him to do for us and therefore we learned how we could proceed in the future with our own studies, with our own research in this area." [industry staff]

"This was the perfect internship for the time because I had just finished up a class on non-linear functional analysis, and a lot of the material we learned in there, I was able to directly apply to this role. So that was just fantastic because some things we treated very abstractly but when I got to see them in an applied setting, it made a lot more sense to see them up close and personal, so to speak." [student]

"Yeah. I liked it a lot. ... making figures, visualizing that data, and also, there was another time where they had to run a model on something to model how the energy system would change, but they had to have some data on how much projections of the amount of CO2 stored underground, and so I went and found that somewhere. I felt like I was useful, definitely. And something that I found, actually, my supervisor thought could be turned into a paper that I could be first author on, so we're finishing up that draft right now, actually." [student]

Recommendations

Industry Partners Recommendations

Industry partners offered recommendations on preparing students for the internships, clarifying some of the INMAS processes, and building more collaborations with industry. Industry staff also had some recommendations for how companies can improve their own preparation for hosting an intern.

Preparation of Students

- ***Increase the awareness and understanding of the challenges of messy, real-life data and the importance of data cleaning***
- ***Provide training in how to present to a non-technical audience***

“Seventeen slides with a bunch of math on it does not resonate with business people. I appreciate it, I respect it, there’s a huge amount of skill, but I assure you that if I go to a bunch of people who have built a very successful career selling chips, that does not resonate with them.”

- ***Provide an introduction to agile methodology***

“The way we manage the team, we use the agile methodology. That's kind of a trend. Nobody even really knows what that means, but there are definitions of agile, so maybe some familiarity with having standups, having a kanban board. We manage the team through a backlog of features or specific scripts or code fixes people have to do and it goes from backlog then to doing, and then, there's an owner of that. And then, when there's blockers, they talk about it. What's blocking me from getting this done? They'll talk about it every day. Then, it eventually moves to done.”

- ***Outline expectations of communication with supervisors/teams and the importance of self-direction***

INMAS Process

- ***Student and University Calendars***

“INMAS is working with different universities and they have different schedules. But I think it may be advantageous to the companies that participate as well as to the interns, if we all knew what the recurring calendar is. In terms of, ‘When do I post? or When should I start reaching out?’”

- ***Review Sheets***

“I felt like there was confusion on those little review sheets that we needed to fill out. I had to fill them out and then I sent it to the intern because, candidly, I want them to be successful. So I had them read it before I sent it and make sure that it all made sense. But then they weren't allowed to send it, I had to send it because it had to come from my email, but it had to have their signature on it. ... Also, I guess the other side of that is if it was a less than stellar review, it could cause a little drama, but that process was a little clunky to me, just given the number of other things and paperwork floating around, that was a bit of a distraction.”

- ***Project Report: More involvement with industry partner***

“The interns had to create a report for INMAS. I think a better practice would've been for us to have them get into that process... Have them get into that routine early on from the start of an internship rather than later on in the internship. So that way they didn't have to dig and think through what they've done. It was having a process where documentation was in an ongoing process and part of their weekly deliverable level there.”

- ***Payments for internship***

“We didn't know how we would get paid, how they would get paid. That wasn't very clear to us, I guess, at the very beginning. After we went through the first few payments for them, we figured out. So that wasn't clear. But now that we know, I think we can be prepared.” [This company was using a grant to pay the interns.]

- ***Let students know about the possibility of extending internship or, alternatively, let companies know when that is or isn't possible.***

“We've talked to the students when we've done this before about potentially working with us part-time to be able to maybe finish up some aspects of the project that didn't get quite done. And that obviously just becomes an issue for them because they're either teaching or doing their own research and just doing that is, it is not very practical for them once they leave us. But being able to tack on an extra month or a few extra weeks because by the time they've learned the ropes and they've dug in and are really starting to make some progress, the program comes to an end.”

Company Process

- ***Have an in-person meeting at the beginning of the internship, especially for those positions that will be remote.***

A couple of companies noted this. One did include the intern for their usual in-person team meeting and another company is considering that step for next year's internship.

“I think it would work, if students are open to spend one week on site to understand what we are actually talking about. ... So would it be appreciated by students if there's opportunity for them to come visit and actually spend a week or so? Obviously, all the expenses will be covered by the company in this case?”

“We had everybody come in for a couple of days just to meet each other and create a network and a sense of cohorts. So that way they can learn from each other. That was the first time we're doing it, but I think it'll work reasonably well.”

- ***Have clearly designed project***

“Just to reiterate, at least from our standpoint, I think the best interactions or best production we've gotten from the program is to have as much face-to-face time with the students as possible. And we have learned, it's really critical to have a well-defined problem for them because of the time and so forth.”

Enhance Collaborations

- ***End-of-the-Year Gathering***

“Some type of a wrap event or something would be nice, where we could kind of see where other students end up.”

- ***Establish an Industry Steering Committee for INMAS***

“A steering committee of industry folks might be a worthwhile consideration for INMAS. A steering committee that can give those perspectives, but be more part of that iterative process of growth.”

- ***Establish company networks similar to the hub idea for universities***

“There could be a INMAS kind of company network too.” [This industry staff person noted that they are still working with the interns who moved to New Mexico and is now local to one of their clients.]

“I know you guys have a group of universities you work with, but we also have affiliations with other universities through their research labs when we do work. How can we connect some of those universities with INMAS? How can you grow, I guess? Or do you plan on growing in terms of having more universities? Because sometimes our federal contracting agency works with a certain university and they want us to work with that lab and we know there are postdocs there or PhD students about to graduate. We work with the University of South Florida, for example. We work with University of Dayton in Ohio. These are some of the universities we work with and there are some good people there that want to find jobs and we just don't have time to absorb them. So I wanted to see if I could do them some benefit by referring them to you.”

Student Recommendations

- ***Increase stipend and have a clearer process for payments***

“Yes, to be totally honest, I think considering the kind of difficult internship these are, research internships and the kind of students to coming in to work on this, the stipend really needs to increase. If you look at an undergrad internship stipend at some financial institute in New York, ours is laughably small.”

“It was marketed as around \$1,000 a week or so. INMAS said, ‘This much total.’ And that total is to \$1,000 a week. And that was always stated as per week. That was what I had been budgeting in my mind is, ‘This much per month, this much per month, the last month.’ And then I was very surprised to hear that it ended up being split up into two lump sums, one half being entirely at the beginning and then one half entirely at the end after your paper was approved. And I definitely feel like that we should have been informed about that. You can make the argument that it's the same net amount of money, and we're not losing out on anything, but rent still needs to be paid and groceries still need to be bought in those weeks in between.”

- ***Expand opportunities for students to learn about industry***

“During the workshops, if they can bring more industry people, the companies and give some presentations over these other opportunities that they have, these are the skills that they're looking for, kind of like that. I feel like it would've been good as well.”

“If you want people to go to the industry, you cannot just say, ‘Okay, when you have gone through 20 years of the education, when you made all of these plans, go to the industry for one month within your whole thesis idea and choose industry after that.’ I think more students should go for the internships at their first year, maybe even their master's degree as well. So departments I think should more be focused on this, introducing students to the industry. Not just having a career fair once in a year type of thing, because all the speakers who comes to university will be other academic people.”

- ***Have more of a structure for companies to respond to resumes and interviews***

“I was really expecting some answers from every place I applied. No matter what. When you apply to some random website, you don't expect a response. But when I applied to the INMAS, I was expecting some response and half of the companies I applied, I think I applied five or six of them. I didn't get any answers. I didn't get any response. I didn't get any contact. Nothing.”

“It would be nice to have some idea where my application stands during the process. Because I started applying maybe February or something, and it wasn't until mid-April that I heard back anything at all. And I don't even know what happened to the remaining of the companies. There was just zero feedback. So it would be nice to get some feedback because graduate students for the summer, they need to decide what they're going to do. Are they going to teach? If they're going to teach, they need to decide early. So either move this entire decision process much earlier or have some more feedback where it stands. If some company doesn't like it, they should let me know as early as possible.”

- ***Start the internship selection process earlier***

“The process of interviewing should maybe start much earlier. Because for example, this year, when I'm applying, I see that a lot of the biggest finance companies, which is often a big area of interest for math graduate students, they already have their internship postings ready even in August. And I've currently applied to a few. So occasionally, some of them reopen their positions in January, which is more suitable for INMAS students, but sometimes they don't.”

- ***Expand the network of participating universities***

“Okay, so I don't know if this is in your request wheelhouse, but one of the places that I applied to was in Southwest Virginia in Roanoke, which is fantastic. ... I would really love to see them become a part of this network. So my recommendation is to reach out to them. I can provide the names. I'm sure somebody would be interested in making partnership with this network.”